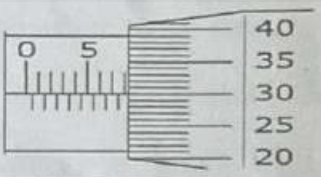
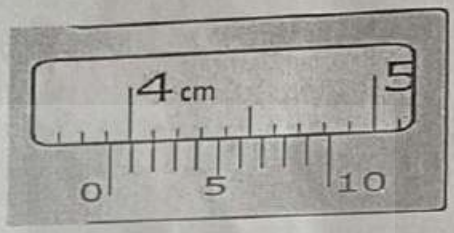
** Circle the correct answer, and move these answer to the table above :(13.5 marks)

1) Using significant figures rules, find the sum of these readings:



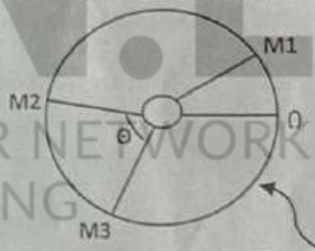




A)51.mm	B) 82.mm	C) 81.5mm	D) 81.50mm

2) Three forces are not in equilibrium, as M_1=65g, M_2=80g and M_3=50g. In order to bring the system in equilibrium, the value of Θ is:

A) 116.7°	B) 125.8°	C) 101.5°	D) 135°



Force table

3) If a student used an elastic ruler with a maximum range (15 cm) to measure the circumference of a circle which has a radius of (2.2m), then the uncertainty in measuring the circumference is:

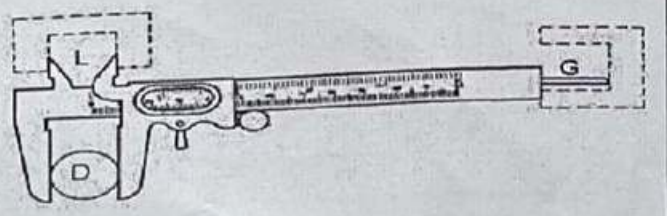
A) 27.6 mm	B) 0.27 mm	C) 22.61 mm	D) 46.05 mm

C.N.E
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4) Three forces are in equilibrium, F_{2x}= 0.25 N , F_{2y}= - 0.4 N, the direction of the resultant between F_1 and F_3(According to positive x-axis) is:

A) 57.99°	B)122.01°	C)302.01 °	D) 237.99°

5) In this picture we can say:



A) G≤D≤L	B) L=G=D	C) L>D>G	D) L<D<G

6) Using (S.F)-rules. The result of the next addition process (2.41 + 10.00) is:

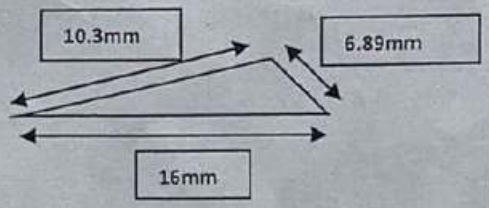
A) 24.1	B)12	C) 12.41	D)12.4

7) Digital balance, when it turned on give us (0.0 g) .Then the error in the value of mass using this balance is:

A) 0.1 g	B) 0.01 g	0.005 g	D) 0.05 g

8) Student used different instruments to calculate the circumference

Of triangle, the error in the circumference is using (S.F) rules:



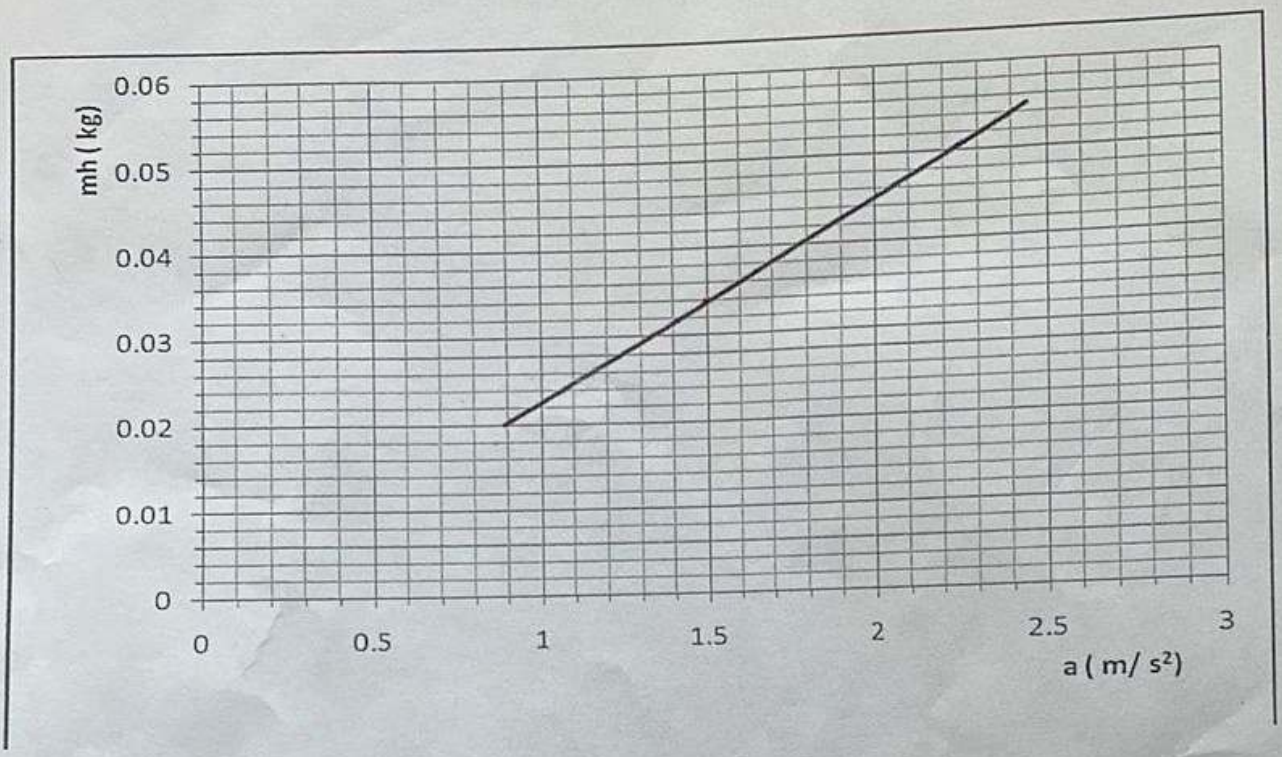
A) 1mm	B) 0.5 mm	C) 0.56 mm	D) 0.555 mm

9) A spherical ball with radius (R), the uncertainty (error) in its surface area is:

A) 6A. ΔR/R	B) A. ΔR/R	C) 2A. ΔR/R	D) 3A. ΔR/R

Q (2): (7.5 marks)

A student drew the graph shown below. In Newton's second law experiment, as (s) and (Mw+mh) are constant in all trials (S=90cm ((where (S) is the distance between the two photo gates)), (Mw+mh=0.480 kg)



Note: Use the last graph only, to answer the next questions.

(1.5 marks) 1) the physical meaning of the slope and find its magnitude:

$slope = \frac{M_w + m_h}{g} = \frac{0.480}{10} = 0.048$

$m_h = \left(\frac{M_w + m_h}{g}\right) a$ (slope)

(1.5 marks) 2) the experimental value of acceleration due to gravity:

From equation $m_h g = (M_w + m_h) a \Rightarrow \frac{a}{g} = \frac{m_h}{M_w + m_h}$ at time = 0.894 = 0.10625

$a = \frac{25}{t^2}$

(1.5 marks) 3) at (t =0.894 s), the force that drives the system:

$F = (M_w + m_h) a$

$= (0.480)\left(\frac{2(0.90)}{(0.894)^2}\right)$

$F = 1.08\ N$ 1/5

(1.5 marks) 4) the mass of the sliding cart M_w at the same previous time (t=0.894s):

From the graph m_h at $t = 0.894$ is $m_h = 0.051$ 1/5

$M_w + m_h = 0.480$

$M_w = 0.480 - 0.051 = 0.429\ kg$

(1.5 marks) 5) One aim of Newton's second law experiment is:

Calculate the experimental value of gravity by $F_g = m_h . g$ and neglecting all force like friction and air resteince

Good luck

To study relationship between Force and accel when mass is constant